The fixed-dimensional lower convexity envelope is zero

Disproof of Conjecture 00000000947

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Abstract

Already in dimension two, the lower envelope of the modulus of convexity over all norms is zero. Every finite ℓ_q^2, $1 < q < \infty$, has strictly positive modulus at positive separation, so a minimizing exponent cannot tend to two as the separation tends to zero. Even if one restricts the envelope to uniformly convex norms, its infimum remains zero for separations below two. Lean verifies actual norm axioms, genuine modulus infima, strict positivity, the exponent-selection contradiction, and the uniformly convex approximation.

The genuine modulus and its lower envelope

Work on the fixed real vector space $E = \mathbb{R}^2$. For any norm N on E and $0 \le \varepsilon \le 2$, use the standard unit-sphere form of Clarkson's modulus [2, Section 1]:

$$P_N(\varepsilon) = \{(x, y) : N(x) = N(y) = 1,\ N(x - y) \ge \varepsilon\},$$
$$\delta_N(\varepsilon) = \inf_{(x,y)\in P_N(\varepsilon)} \left(1 - N\left(\frac{x+y}{2}\right)\right).$$

This is equivalent to the unit-ball version in this dimension; the cited section recalls the equivalent conventions. The infimum here is over all admissible unit pairs, not just symmetric test vectors. It is nonempty: for any nonzero vector e, the pair $x = e/N(e)$ and $-x$ has separation two. The triangle inequality makes every displayed defect nonnegative. Thus $\delta_N(\varepsilon) \ge 0$. Let $L(\varepsilon) = \inf_N \delta_N(\varepsilon)$, over all genuine norms on this fixed plane.

For the actual sup norm $N_\infty(s, t) = \max(|s|, |t|)$, take

$$u = (1, \varepsilon/2), \qquad v = (1, -\varepsilon/2).$$

Both have norm one, their distance is ε, and their midpoint has norm one. Their defect is zero. Together with nonnegativity this proves

$$\delta_{\ell_\infty^2}(\varepsilon) = 0, \qquad L(\varepsilon) = 0 \quad (0 \le \varepsilon \le 2).$$

Positive modulus for every finite exponent greater than one

For real $1 < q < \infty$, use the genuine norm

$$N_q(x) = (|x_0|^q + |x_1|^q)^{1/q}.$$

We prove positivity of its actual modulus without assuming an explicit Hanner formula. For real a, b, convexity and the absolute-value triangle inequality give

$$\left|\frac{a+b}{2}\right|^q \le \frac{|a|^q + |b|^q}{2}.$$

The inequality is strict when $a \neq b$. If $|a| \neq |b|$, strict convexity of $t \mapsto t^q$ on $[0, \infty)$ gives strictness after the triangle inequality. If $|a| = |b|$ but $a \neq b$, then $a = -b \neq 0$, and strictness is immediate. Consequently distinct N_q-unit vectors satisfy

$$N_q\left(\frac{x+y}{2}\right)^q < \frac{N_q(x)^q + N_q(y)^q}{2} = 1.$$

Fix $0 < \varepsilon \le 2$. The admissible set $P_{N_q}(\varepsilon)$ is nonempty and closed. Each coordinate of each unit vector has absolute value at most one, so this set is bounded in $\mathbb{R}^4$, hence compact. The continuous defect attains its minimum there. Its value is strictly positive at every admissible pair, since the positive separation forces $x \neq y$. Thus the attained minimum is positive:

$$\delta_{\ell_q^2}(\varepsilon) > 0 \qquad (1 < q < \infty,\ 0 < \varepsilon \le 2).$$

Compactness is essential here; pointwise positivity alone would not justify a positive infimum.

The same conclusions also follow directly for the unit-ball convention $N_q(x), N_q(y) \le 1$. Its admissible set is again compact. If either norm is below one, the triangle inequality makes the midpoint norm strictly less than one; if both are one, use the strict inequality above. Thus its attained minimum is positive as well. The displayed sup-norm pair still gives exact zero.

Theorem 1. *No choice of minimizing exponents $q(\varepsilon)$ for the fixed-dimensional lower envelope can satisfy $q(\varepsilon) \to 2$ as $\varepsilon \to 0^+$.*

Proof. Such convergence forces $1 < q(\varepsilon) < \infty$ for all sufficiently small positive ε. At any such $\varepsilon < 2$, its modulus is strictly positive, while the lower envelope is zero. Therefore it does not attain that envelope. □

This contradicts the source's asserted small-separation selection already for dimension two. No assertion about the separate large-separation limit or endpoint slopes is needed to refute the stated conjunction.

Restricting to uniformly convex norms does not repair it

The exact source imposes no uniformly convex norm restriction. Nevertheless, the same obstruction holds on that smaller class for $0 < \varepsilon < 2$.

The positive modulus just proved shows that every finite ℓ_q^2, $q > 1$, is uniformly convex. For such a q put $k_q = 2^{1/q}$ and take the actual unit vectors

$$x_q = \frac{(1,1)}{k_q}, \qquad y_q = \frac{(1,-1)}{k_q}.$$

Their separation is $2/k_q$ and their midpoint norm is $1/k_q$. Hence, whenever $\varepsilon \le 2/k_q$, the genuine modulus satisfies the *upper bound*

$$0 < \delta_{\ell_q^2}(\varepsilon) \le 1 - 2^{-1/q}.$$

We do not assign this trial-pair expression as an exact modulus formula.

As $q \to \infty$, $2^{-1/q} \to 1$. Given any fixed $0 < \varepsilon < 2$ and any $\eta > 0$, choose a finite $q \ge 2$ large enough that

$$2 \cdot 2^{-1/q} > \varepsilon, \qquad 1 - 2^{-1/q} < \eta.$$

This produces a genuinely uniformly convex norm with $0 < \delta_{\ell_q^2}(\varepsilon) < \eta$. Since every norm modulus is nonnegative, even the infimum over only uniformly convex norms is zero. Every finite $q > 1$ lies strictly above it, so the claimed minimizing selection tending to two remains impossible. This approximation assertion deliberately excludes $\varepsilon = 2$; no conclusion about that endpoint is needed.

Actual geometric definitions in Lean

The underlying vector space is `Fin 2` to $\mathbb{R}$. The predicate `IsNorm` consists exactly of nonnegativity, positive definiteness, the triangle inequality and absolute homogeneity. The definitions `modulus` and `envelope` are actual infima over all admissible unit pairs and all such norms. Domain nonemptiness is proved for every genuine norm, so no empty-infimum default produces the zero envelope.

Finite ℓ_q norms are the actual Mathlib `PiLp` norms, pulled back by the canonical coordinate equivalence. Their sum-of-real-powers formula, continuity, strict midpoint inequality and compact admissible sets are proved. A general uniformly convex instance is derived from these facts, rather than assumed as an unavailable library instance.

Formal conclusions and reproducibility

The principal theorems are:

- `pairs_nonempty` and `envelope_zero`: all norm moduli are properly defined on the domain and the exact envelope is zero.
- `lp_modulus_pos` and `no_finite_lp_minimizer`: every finite $1 < q < \infty$ has positive actual modulus and cannot attain the zero envelope.
- `no_selector_tending_two`: direct negation of an arbitrary minimizing exponent selector whose limit as $\varepsilon \to 0^+$ is two.
- `lp_uniformConvex` and `arbitrarily_small_uniformly_convex_lp`: actual Mathlib uniform-convexity certificates and arbitrarily small positive moduli in that class.

The selector theorem permits extended exponents as well: convergence of their real values to two forces them eventually into the finite range $q > 1$. The approximation theorem gives a real exponent and an actual uniformly convex `PiLp` space, together with strict $0 < \delta < \eta$, for every fixed $0 < \varepsilon < 2$.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From the submitted `lean/` directory:

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The entire project is built in a fresh directory. The strict check treats warnings as errors, and the eight principal axiom audits contain only standard foundational axioms. There are no proof placeholders, custom axioms, numerical approximations or external computation oracles. No auxiliary program is required. Compile the document with `tectonic main.tex`.

The exact bilingual source, provenance, actual build logs, artifact hashes and PDF verification records are included. An agent developed the proof and supplied an adversarial self-review. The parent separately reviewed the complete proof, its scope, verification evidence and every final PDF page. No external independent review or novelty is claimed for these elementary convexity facts.

References

[1] The Last Math Competition, Conjecture 00000000947. Exact bilingual source and provenance included.

[2] E. Llorens Fuster, *Moduli and Constants*, April 14, 2006, Section 1. Author-hosted survey. The citation supplies standard terminology and equivalent modulus conventions. All infimum and strict-positivity arguments used in this disproof are given above.